EEE96005. ARTIFICIAL INTELLIGENCE

1. This question is about declarative programming and Prolog.

   Important: any answer to any part that doesn't use comments to illustrate how the predicate is queried, and to explain adequately what each line of code does, will get zero marks.

   a) Given two lists of unordered elements, of unknown and possibly different lengths, write a Prolog predicate that performs a pairwise comparison of the elements of the list, and counts the number of pairwise matches.

   For example the pairwise comparison of `[1,2,3,4]` with `[1,2,3,5,7,9]` returns 3, as does the pairwise comparison of `[1,2,3,5,7,9]` with `[1,2,3,4]`.

   [6]

   b) Given two lists of different lengths, one known to be shorter than the other, write a Prolog predicate that computes the largest pairwise comparison of the shorter list with a sub-list of the longer list.

   For example, the best pairwise comparison of `[1,2,3,4]` with `[7,1,2,3,7]` is 3.

   [4]

   c) Given a list of ordered elements, write a Prolog predicate that computes the set of pairwise preferences over the list.

   For example, for the list `[x,y,z]`, the set of pairwise preferences is: `{(x,y),(x,z),(y,z)}`.

   [4]

   d) A matrix in Prolog can be represented using a list of lists; for example the 2x2 identity matrix can be represented as `[ [1, 0], [0, 1] ]`.

   Write a Prolog predicate that generates the identity matrix for a matrix of size *N*.

   [6].

2. This question is about search algorithms and the General Graph Search Engine (GGSE).

a) Explain, with the help of an example, how a state space can be represented in Prolog, so that it can be used for planning and problem-solving with the GGSE.

[4]

b) There are two professors, three students, and a boat on one bank of a river. The boat can hold at most one professor or (exclusively) two students, and at least one person must be in the boat to make it move. They all want to get to the opposite bank.

Formulate this river-crossing problem as a state space, so that it could be solved with the GGSE.

[6]

c) Compare and contrast breadth first, depth first, uniform cost, Beam, best first and A* search algorithms according to four criteria for evaluating different search algorithms.

[3]

d) Give, in Prolog, an extension of the GGSE for iterative-widening, iterative-deepening Beam search. Assume, on each iteration, that both the beam width and the depth limit are incremented.

Briefly explain how a path is selected for expansion (`choose/3`) and how a 'larger' graph is constructed (`add_to_paths/3`) for recursive search.

[5]

e) State, with justification, for what kind of expected search space this algorithm might be suitable.

[2]

3. This question is about graph representation, the A* algorithm and heuristics.

a) Given a graph explicitly defined as $G_S = \langle N, E, R \rangle$, give an inductive definition of the paths in the graph rooted on a node $S \in N$.

[2]

b) Given a graph explicitly defined as $G' = \langle S, op \rangle$, give an inductive definition of the paths in the graph.

[2]

c) State what condition needs to hold for the paths produced by a graph G_S to be the same as the paths produced by G'.

[2]

d) Explain how the A* search algorithm expands the paths in a graph defined as $G' = \langle S, op \rangle$.

Prove that the solution path returned by the A* search algorithm is optimal.

[4]

e) In the context of A* search, explain the conditions for a heuristic to be usable and admissible.

Explain why these conditions might make one heuristic perform "better" than another.

Comment on this explanation with respect to optimality and complexity.

[6]

f) Explain how heuristics can be used in search algorithms for 2-player games.

[4]

4. This question is about knowledge representation and reasoning with **KE**. Proofs in **KE** should be properly annotated to show the inference steps, i.e. showing the inference rule used and line number(s) of the premise(s),

a) Consider the following statements:

1. *Every maximal being is more-provable than every impossible world.*
2. *There is an inter-dimensional travel method that is more-provable than every mathematician.*
3. *Every inter-dimensional travel method is an impossible world.*
4. *More-provable is transitive.*
5. *Thör is a maximal being.*
6. *Gödel is a mathematician.*

i) Express these statements as formulas of Predicate Logic.

ii) Convert these formulas into Horn Clauses.

iii) Using resolution and showing all unifiers, prove by negation that:

Thör is more-provable than Gödel

iv) Convert the Horn Clauses into a Prolog program.

[8]

b) Consider the following argument:

If spring is sprung and the grass is green, then either I wonder where the birds are or I see that the birds are on the wing.
It is not the case that if the grass is green, then I wonder where the birds are.
Therefore, if I do not see that the birds are on the wing, then spring is not sprung.

Rewrite this argument in propositional logic, then use the calculus **KE** to prove that the argument is valid.

[6]

c) In the *Obligatio Game*, a finite number n of rounds is chosen, depending on the severity of the exam. The examiner then gives the candidate successive assertions, $\phi_1, \phi_2, \ldots, \phi_n$ that she has to either *accept* or *reject* as each one is put forward. In the former case, ϕ_i is added to the candidate's stock of commitments; in the latter case, the negation $\neg\phi_i$ is added.

The candidate passes the exam if she maintains the consistency of her stock of commitments throughout all n assertions.

Suppose the candidate is exposed (successively) to the following three ($n = 3$) assertions:

1. $p \rightarrow (q \vee r)$
2. $p \rightarrow \neg s$
3. $s \leftrightarrow q$

Suppose the candidate channels her inner Nike and accepts every premise: should she accept or reject the assertion $p \rightarrow r$, or doesn't it matter?

(Hint: use **KE** for model building. If the proof has at least one open branch, then this implies a consistent sequence of answers.)

[6]

5. This question is about **KE**, Modal Logic and the Event Calculus. **KE**-trees (proofs) should be properly annotated to show the inference steps, i.e. showing the inference rule used, line number(s) of the premise(s), and the label for the world.

a) Using the Calculus **KE** for Modal Logic, prove that the following formula is valid in the logic **K**.

$$\Diamond(A \wedge B) \rightarrow (\Diamond A \wedge \Diamond B)$$

Using the Calculus **KE** for Modal Logic, prove that the following formula is *not* valid in the logic **K**.

$$(\Diamond A \wedge \Diamond B) \rightarrow \Diamond(A \wedge B)$$

Construct a Kripke Model as a counter-example and use it to explain the proof.

Explain whether or not the same results hold for logic **S5**.

[10]

b) Explain, with the use of an example, how knowledge is represented and reasoned with in:

i) state-space formulation as a graph;

ii) a Kripke model; and

iii) axioms of the Event Calculus.

[10]